

Abstract

In our universe, the collisions of extremely dense objects, such as neutron stars, create ripples through space. These ripples are gravitational waves that expand and compress the space they travel through. We can detect these waves using interferometers, like the LIGO detectors. However, gravitational waves have intrinsic uncertainties between the true waveform we detect from space and the waveform models, or waveform approximants, we use for parameter estimation. These uncertainties can be quantified in amplitude and phase, δA and $\delta \phi$. Our goal is to use machine learning to pick out the differences between two approximants using the uncertainties between them. This requires a large set of training data to be given to the program. Training data provides information in the machine learning process, helping us further with our analysis. Our training data utilizes GW170817 sample sets and includes: a range of frequencies relevant to gravitational wave detection, neutron star source parameters, amplitude uncertainty, and phase uncertainty for each waveform. We needed to reduce the size of the waveform uncertainty data to make training a machine learning algorithm feasible. We wanted each uncertainty to be expressed in 10 parameters. This was achieved utilizing a series of Chebyshev polynomials to allow for a concise parameterization of waveform uncertainty within the tolerance of uncertainties in the LIGO detectors.

Methods

To train a machine to identify the differences between the approximants, we need to feed it a large quantity of samples. For our purposes, 10,000-15,000 sample sets would be required. These sample sets needed to give the program all relevant information about the waveforms, such as the frequencies, the neutron star source parameters, and the waveform uncertainties. Using this data, The machine learning algorithm would learn to choose the waveform uncertainty that fits the data.

The training data needed to include a fixed frequency grid, which we defined to be a geometric progression between 20.0 Hz and 2028.0 Hz. This grid contained 500 points and would be used to interpolate the waveforms. We then needed to find the waveform uncertainties, δA and $\delta \phi$.

To find δA and $\delta \phi$, we first need to generate two frequency-domain waveforms across our frequency grid. These waveforms were generated with the same neutron star source parameters, but two different waveform approximants:

IMRPhenomPv2_NRTidalv2 and SEOBNRv4T_surrogate

These are two state-of-the-art models used for merging neutron stars. The waveforms predict two waveforms, W_1 and W_2 , which are expressed as sets of strain values, which are complex numbers in the form $a + bi$.

The waveform uncertainties were calculated point by point with the following equations, where s_1 and s_2 are strain values corresponding to W_1 and W_2 :

$$s_1 = a_1 + b_1 i$$

$$s_2 = a_2 + b_2 i$$

$$\delta A = \frac{|s_1| - |s_2|}{|s_1|}$$

$$\delta \phi = \arctan\left(\frac{b_1}{a_1}\right) - \arctan\left(\frac{b_2}{a_2}\right)$$

$\delta \phi$ can drift between the models from an overall shift in time and phase, which would not be detectable for parameter estimation. To remove this effect, we fit a line to $\delta \phi$, weighted by the power spectral densities corresponding to the GW170817 event¹. We then subtract this line from the data, removing the overall shift in time and phase. This leaves us with residual phase uncertainty, or $\delta \phi_R$.

Optimizing the Uncertainties

Because the waveforms were generated across our 500-point frequency grid, the waveform uncertainties were expressed as a series of 500 points. Spanning over 10,000 sample sets, running through 500 points of δA and 500 points of $\delta \phi_R$ for each waveform would slow the process down far too much to be considered useful. We had to find a way to reduce the size of δA and $\delta \phi_R$. One would likely see the solution as simply reducing the number of points. This, however, would not work as there would be no way of recovering the data lost between the new points. Minimizing this loss would mean minimizing the reduction of points. In turn, the reduction of size would not be efficient. The other solution would be to attempt to fit a function to the uncertainties, an option explored in Figure 1.

δA and $\delta \phi_R$ as Functions of Frequency

$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_2(x) = 2x^2 - 1$$

$$T_3(x) = 4x^3 - 3x$$

$$T_4(x) = 8x^4 - 8x^2 + 1$$

$$T_5(x) = 16x^5 - 20x^3 + 5x$$

$$T_6(x) = 32x^6 - 48x^4 + 18x^2 - 1$$

$$T_7(x) = 64x^7 - 112x^5 + 56x^3 - 7x$$

$$T_8(x) = 128x^8 - 256x^6 + 160x^4 - 32x^2 + 1$$

$$T_9(x) = 256x^9 - 576x^7 + 432x^5 - 120x^3 + 9x$$

$$\delta A(f) \sim \sum_{i=0}^{N-1} a_i T_i(f)$$

$$\delta \phi_R(f) \sim \sum_{i=0}^{N-1} b_i T_i(f)$$

$[a_0, \dots, a_{N-1}]$ and $[b_0, \dots, b_{N-1}]$ would be the sets of parameters that represent the waveform uncertainties, where N is the number of parameters wanted.

Figure 1. The first 10 Chebyshev polynomials². The curve-fits are expressed as a series of these polynomials, each multiplied by a constant, a_i and b_i . These N constants would be the parameters that define the parameterized waveform uncertainties.

Testing proved that this method of using series of Chebyshev polynomials produced accurate representations of the waveform uncertainties given enough parameters, or a large enough N. To get a near perfect fit, upwards of 25 terms in the function were needed, but this was not as efficient as it could be. Ideally, the number of parameters needed to express the data would have to be less than or equal to ten to fully optimize the process. To reach this goal, an error margin needed to be defined.

How good do the fits need to be?

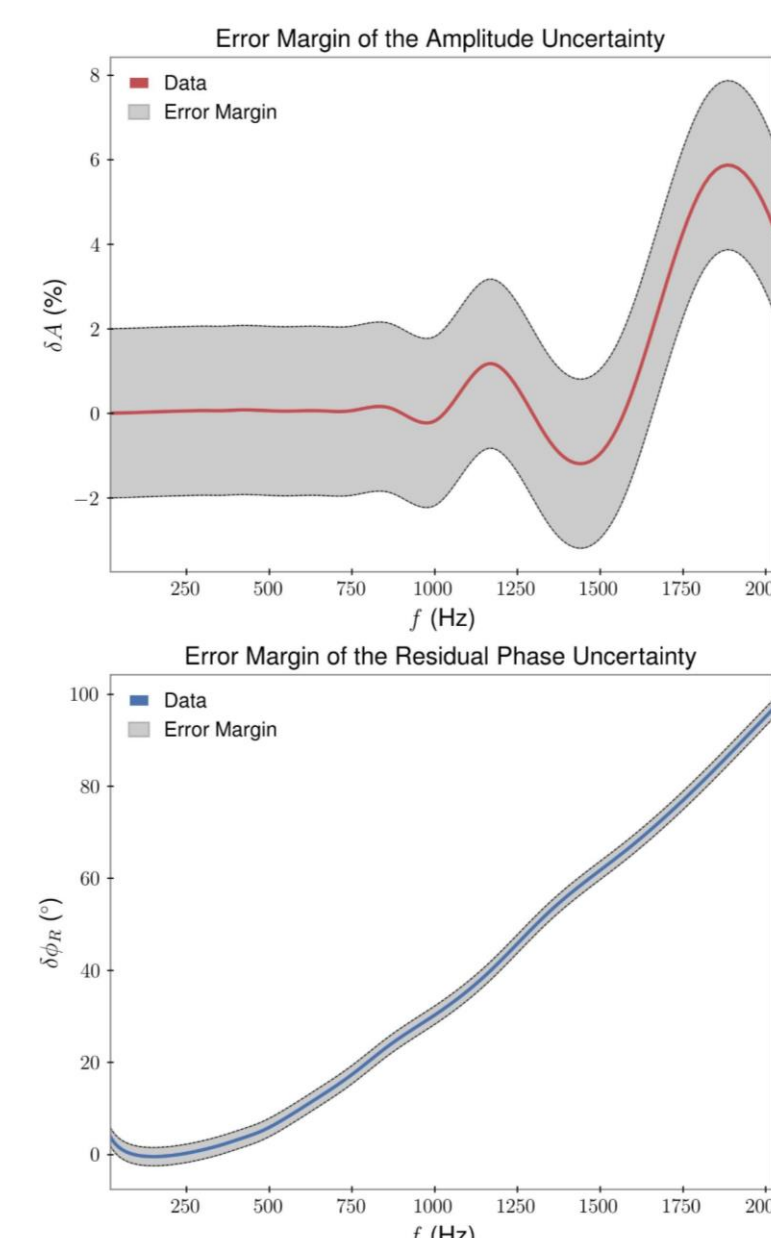


Figure 2. Example of each waveform uncertainty, δA and $\delta \phi_R$. Using the LIGO detector uncertainties, an error margin was defined, shown as the gray-shaded area. This margin for $\delta \phi_R$ is far stricter than that of δA , which forced the number of fit parameters to be exactly 10 rather than below it. δA itself can be fit in 6 parameters.

The LIGO detectors, as with all equipment, have some uncertainties in their measurement. During the O3a observation period, these uncertainties were³:

2% Uncertainty in Amplitude
2° Uncertainty in Phase

Using these detector uncertainties, we were able to incorporate error margins into the waveform uncertainties, as shown in Figure 2.

If the curve fit stayed within this margin, it was accepted as a proper representation of the data. This allowed our number of fit parameters to be decreased substantially, as the accuracy was not as important.

With the error margins defined, the number of parameters needed to represent the data dropped to ten, which was within the margin we wanted. Our parameterized uncertainties then became of the following forms:

$$\delta A(f) \sim a_0 T_0(f) + a_1 T_1(f) + \dots + a_8 T_8(f) + a_9 T_9(f)$$

$$\delta \phi_R(f) \sim b_0 T_0(f) + b_1 T_1(f) + \dots + b_8 T_8(f) + b_9 T_9(f)$$

The sets $[a_0, \dots, a_9]$ and $[b_0, \dots, b_9]$ then became the parameters used to express δA and $\delta \phi_R$ in the data file. Because of the simple form of the formulas, recreating the data from the parameters was proven to be an easy task given the proper code. We now had all the pieces of the training data together: the frequency grid, the neutron star source parameters, and the parameterized waveform uncertainties. The data file itself would then be created.

Conclusion and Future Work

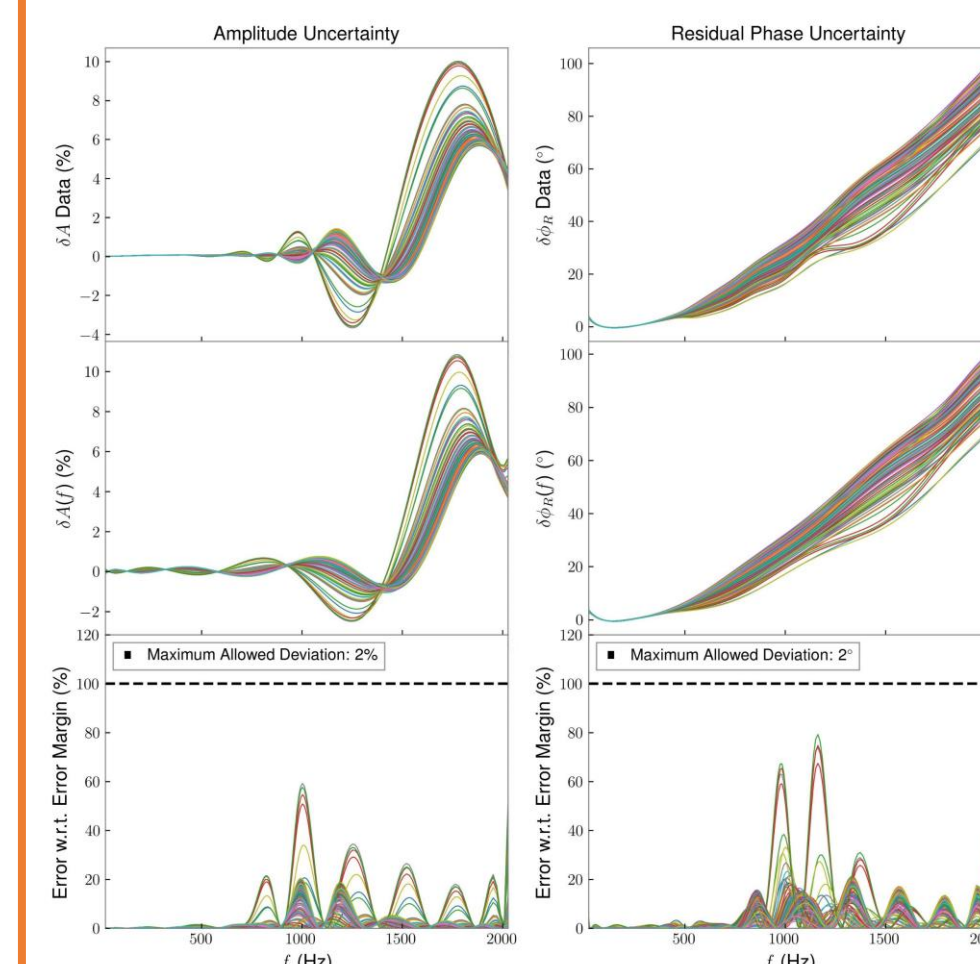


Figure 3. 100 draws of the waveform uncertainties, their parameterized representations, and the errors between them relative to the error margin. Despite the visual differences between the data and their representations, the errors between the two are well within the acceptable margin.

The parameterization of waveform uncertainty for use in machine learning has proven to be an effective way to represent the data in a small quantity of information. By approximating δA and $\delta \phi_R$ with a series Chebyshev polynomials, the 500 data points were reduced to 10 parameters. To achieve this, we utilized the systematic uncertainties in the LIGO detectors to define an error margin, allowing for further optimization in the parameterization process.

The Training Data

The training data, containing the frequency grid, neutron star source parameters, and the parameterized waveform uncertainties, can be automatically generated using a program coded in Python. The program loops through its code over 10,000 times. In each loop: two waveforms are generated, the uncertainties are calculated, the uncertainties are parameterized, and the data is stored, with each loop taking about 0.25 seconds to complete.

The data file contains over 10,000 rows. Each row contains the data presented in Figure 4, showing row number 71 as an example. The neutron star source parameters are taken from the parameters used to generate the waveforms. Those we present in the data file can be chosen by the recipient of the training data. Currently, we choose chirp mass ($\mathcal{M}_c$) and effective spin (χ_{eff}).

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Row number: 71

Frequency Grid: 20.015625 ... 2027.984375; 500 items

Parameters ( $\chi_{\text{eff}}$ ,  $\mathcal{M}_c$ ): [0.02310524 1.18513996]

dA Parameters: -0.003576245715618931 ... 2.75513237254117e-30; 10 items

dphi Parameters: 0.10288482754911091 ... -1.0607733645774758e-29; 10 items

Fit Error as a Percentage of the Error Margin (dA, dphi): [13.03523824 23.74485028]
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Figure 4. One row of the training data set. Each row displays the frequency grid, the chosen neutron star source parameters, the fit parameters for δA , the fit parameters for $\delta \phi_R$ and the error.

Currently, this program does not support tidal deformabilities, notated in neutron star source parameters as λ_1 and λ_2 . These tidal parameters are defining features of neutron stars, so their absence is a major flaw in this program. Writing this program on another software that does support these tidal parameters is ongoing. The use of this training data in machine learning is also ongoing; no results have been reported yet.

References

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